DSFI-CD: Diffusion-Guided Spatial-Frequency-Domain Information Interaction for Remote Sensing Image Change Detection

Xiaolong Li, Yonghui Tan, Kunming Liu, Xuan Wang, and Xusheng Zhou

***Abstract*— The change detection (CD) is vital in remote sensing (RS) image interpretation applications and has been well-improved driven by deep learning (DL). However, existing DL-based methods have poor model robustness due to the effect of a limited number of change samples in the existing public datasets, and on the other hand, existing most CD methods only process the input image in the spatial domain, and insufficiently process it at the frequency-domain level, which leads to its ability to obtain only shallow semantic information of the image, and insufficient ability to deal with the high-frequency information (edges and details) in the RS. In order to address the issue, this article proposed diffusion-guided spatial–frequency-domain information interaction RS CD (DSFI-CD) method, which generates part of the pseudoimages by using a conditional denoising diffusion probability model (DDPM). Furthermore, it suggests a module called spatial–frequency interaction (SFI), which combines the extraction and fusion of the corresponding spatial- and frequency-domain data with the intersection of the dual time–phase features prior to and following the change to give the model more reliable information. In addition, we designed the edge-enhanced (EE) module to further decompose the extracted features in the SFI module into boundary and body information for the change region edge information. We do this by using the concept of the flow field, and we also use the Laplace pyramid and multilevel feature fusion to increase the sensitivity of the model to the boundary while reducing the impact of noise in the nonchange region. Comprehensive experiments on four publicly accessible CD datasets show that DSFI-CD performs better than other state-of-the-art (SOTA) methods.**

***Index Terms*— Change detection (CD), diffusion model, edge enhance, fast Fourier transform (FFT), high-resolution remote sensing (RS) images, information interaction.**

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I. Introduction

CHANGE detection (CD) is a critical task in the intelligent interpretation of remote sensing (RS) images, which aims to detect differences between RS images from different time periods at the same geographical location [1]. Since the Earth observation technology has gradually matured, a multitude of modal RS images has become available for RSCD tasks. Of these, very HR (VHR) RS images have emerged as the most popular data source for RSCD because of their HR, which allows for the acquisition of more detailed information about ground objects. Furthermore, the usage of RSCD tasks in other industries, including urban land use analysis [2], [3], [4], building CD [5], [6], [7], deforestation monitoring [8], [9], [10], and natural catastrophe assessment [11], [12], [13], has been facilitated by the growing volume of RS image data.

RSCD tasks fall into two main categories: classical-based tasks, which were more frequent prior to the widespread adoption of GPUs, and deep learning (DL)-based tasks. Algebra-based, transformation-based, and classification-based approaches have been widely applied in RSCD tasks that follow conventional methodologies. The main methods used in algebra-based approaches to RSCD tasks include image rationing [14], image differencing [15], image regression [16], and image adaptive region measuring [17], [18]. Transformation-based RSCD tasks analyze change, such as wavelet transformation [19], principal component analysis (PCA) [20], Fourier transform [21], and multiscale morphological transform [22]. The RSCD task makes use of classification-based methods, such as postclassification comparison [23], decision tree [24], support vector machine (SVM) [25], [26], random forest (RF) [27], k-nearest neighbors (kNNs) [28], and postprocessing [29]. Although the approaches described above have previously performed well, they are constrained by empirical design and thus difficult to deal with the increasingly rich texture and semantic information in RS images.

With the rapid iteration of computer numerical power and the release of numerous publicly available RS datasets, DL-based methods have become the mainstream for RSCD tasks. As DL computer vision researchers have been diligently developing, similar models—such as ResNet [30], FCN [31], U-Net [32], and others—for extracting deep semantic information from images have been suggested one after another.

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Daudt et al. [33] first used FCN for the RSCD task and proposed three models: FC-EF, FC-Siam-Conc, and FC-Siam-Diff. Peng et al. [34] extended U-Net++ to accommodate dual time–phase inputs in order to facilitate the RSCD task. To address the feature fusion challenge in the RSCD task, Zhang et al. [35] presented a deeply supervised (DS) image fusion network (IFN). Liu et al. [36] presented DTCDSCN to address the sample imbalance problem in the RSCD task. Apart from the previously mentioned CNN-based methods, several researchers have developed related models, motivated by Transfomer [37], to tackle the issue of inadequate ability of capturing long-range dependencies in the RSCD task. BIT [38] effectively models contextual information in the spatial–temporal domain by transforming inputs into a set of high-level semantic tokens via the use of Transformer. Furthermore, to depict long-range details at many scales efficiently, ChangeFormer [39] integrates a multilayer perceptron (MLP) decoder and a hierarchical Transformer encoder in a Siamese network framework. TransUNetCD [40] combines Transformer's strengths with U-Net architecture features, with the goal of overcoming the limitations of purely convolutional networks in learning global contexts and spatial relationships over long distances, as well as compensating for Transformer's lack of localized detail. Meanwhile, with the emergence of some new techniques, such as segment anything model (SAM) [41] as well as Mamba [42], [43], related works have designed corresponding networks based on them to achieve better detection performance [44], [45].

The DL-based approaches discussed above are classified as CNN-based or Transformer-based, and while they all reach state-of-the-art (SOTA) performance in the respective cases, they each have their own set of limitations. First, CNN-based methods suffer from poor detection accuracy due to their limited local receptive fields and sensitivity to spatial fluctuations, as evidenced by the fact that the results are satisfactory in localized areas when carrying out the CD task, but unsatisfactory results are obtained when confronted with larger spatial scale images [46], [47]. Second, while the Transformer-based approach excels in global modeling capability, in RS imagery, the self-attention mechanism in Transformers is inherently insensitive to positional information in input sequences, necessitating enhancements such as positional encoding. In SegFormer, convolutional operations implicitly capture positional information, further improving the Transformer's performance in RS imagery. However, Transformers still face challenges with computational complexity when handling long sequences, which can become a bottleneck in high-resolution and large-scale RS imagery applications. Furthermore, in addition to the aforementioned shortcomings, we believe that the above models currently perform the RSCD task only in the spatial domain of the RS image, and while the model's detection accuracy for texture, details, and so on improves with increased model depth, few models process the frequency-domain information, which is richer in semantic information.

To address the aforementioned issues, we propose a diffusion-guided spatial–frequency-domain information interaction RS CD (DSFI-CD) method, in which the relevant module uses fast Fourier transform (FFT) to extract the image's frequency-domain and spatial-domain information from the feature map, and then fuses the two domains using information interaction. Meanwhile, the RSCD work includes less samples than other RS image interpretation tasks including scene identification, target detection, and semantic segmentation. As a result, the conditional denoising diffusion probability model (DDPM) is used in DSFI-CD to generate pseudoimages with more semantic information from the actual RS images, hence increasing the number of samples in the RSCD task. To achieve sharper boundary identification in the CD task, we created a boundary improvement module that uses techniques, such as multilevel feature fusion and the Laplace pyramid transform (LPT) to gather more comprehensive edge information about the change region.

The main contributions of this study are summarized as follows.

1) In this work, we present a novel method for CD task, called DSFI-CD, that uses conditional DDPM to improve the semantic information in the original RS image while implementing the interaction of spatial- and frequency-domain information in the feature map. In the RSCD task, DSFI-CD collects semantic information more effectively than previous methods.
2) The JD module processes pseudoimages by extracting regions of interest using the overall feature alignment module (OFAM), filtering noise with AdapFilter, and fusing with the main branch through the dual-fusion (DF) module, which enhances the backbone's ability to extract semantic information.
3) The spatial–frequency interaction (SFI) module in the decoder processes various information domains from backbone's feature maps, including flow field alignment and the Fourier information interaction (FII) module. This enhances the model's perception of multidomain input.
4) The edge-enhanced (EE) module in the DSFI-CD decoder is designed to improve boundary extraction in the CD task by incorporating ideas from LPT and multilevel feature fusion.

The remainder of the study is organized as follows. Section II presents existing related work. Section III details our suggested method in depth. Section IV proves the usefulness of our proposed method through experiments. Section V summarizes this article's work.

II. Related Works

In this section, an overview of existing RSCD techniques, including CNN-based and Transformer-based methods, and diffusion-based is offered, along with their benefits and drawbacks.

A. CNN-Based RSCD

With the rapid computer arithmetic iteration and the release of many public datasets in the last decade, the related technologies in the field of computer vision have been rapidly developed. Some general-purpose vision models, such as

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VGG [48], ResNet [30], FCN [31], and U-Net [32], have been widely used in cross-cutting areas, such as medical image processing [49], surface defect detection [50], and RS image interpretation [51] with the development of DL technology.

The DL-based model for the CD task was introduced later than for other RS interpretation tasks. This is due in part to the fact that the CD task requires processing of multiple temporal phases of RS images, which makes it tough to simply transfer the model from the general computer vision domain to the RS domain; on the other hand, the datasets for CD is more difficult to produce than for other RS interpretation tasks because it must be guaranteed that all the images in different temporal phases at the same geographic location at different time intervals have relatively good lighting, cloud cover, and other conditions. For the first time, Daudt et al. [33] applied Siamese strategies and fully convolutional networks to the CD task. They proposed three models, FC-EF, FC-Siam-Conc, and FC-Siam-Di, based on different strategies. FC-EF is built directly from the U-Net model, while FC-Siam-Conc is a Siamese extension of FC-EF. FC-Siam-Di is similar to FC-Siam-conc in that it uses the Siamese network structure, but handles skip connections differently. Peng et al. [34] introduced an end-to-end U-Net++ [52]-based RSCD model that achieves deep supervision via the multiple side output fusion (MSOF) technique, which efficiently combines multiscale feature maps at various semantic levels to build the final CD map. To deal with computing resource needs and model long-distance spatiotemporal interaction in DARNet, a cross-attention mechanism (ESTAM) replaces the standard self-attention mechanism. Shi et al. [53] proposed a DS attention metrics learning network (DSAMNet) to deal with the effects of external factors in CD work, such as illumination, noise, and scale variations, which result in pseudochanges and the problem of detecting noise in maps, as well as to achieve more accurate results. In addition, the researchers constructed the SYSU-CD dataset to address the constraints of RSCD datasets encountered by data-driven DL systems. To address the difficulties in detecting changes in high-resolution RS images due to intricate texture features and fine image details, Zhang et al. [35] proposed a depth-supervised IFM. This network uses the attention module to efficiently fuse the original image depth features and the image disparity features, assisting in the reconstruction of the CD map, while also performing CD through the disparity discriminative network.

Apart from the approaches described above, several more approaches simulate the attention mechanism by using appropriate convolutional modules. Han et al. [54] presented a hierarchical attention network dubbed HANet to address the issue of pixel category imbalance in CD of high-resolution dual-temporal RS images. The goal of this model is to enhance CD performance by merging fine detail and multiscale information. To increase the accuracy of change recognition in VHR dual-temporal RS images, Li et al. [55] proposed DARNet, which attempts to address the inadequacies of existing DL-based CD models while dealing with scale change and pseudochange difficulties. In order to address the issue of the significant discrepancy in recognition accuracy between the main stem and the edge of a changing region, Tan et al. [56] proposed BD-MSA, which simultaneously aggregates global and local information in both channel and spatial dimensions and adapts to feature information at different scales. Ma et al. [57] presented STNet, an RSCD network based on spatial and temporal feature fusion (TFF), which highlights change regions using a temporal and spatial feature fusion module, respectively. Liu et al. [36] introduced a dual-task restricted deep twin convolutional network model (DTCDSCN) that combines CD and semantic segmentation tasks to improve CD performance by learning semantic segmentation of buildings.

Although the above approaches based on CNN have achieved good results, particularly in the border of change region, their limited receptive fields mean that different methods listed above are unable to capture long-range dependencies in RS images well.

B. Transformer-Based RSCD

With the development of the self-attention mechanism [37], computer vision researchers transferred it from the field of natural language processing to the field of computer vision, proposed some models such as Vision Transformer (ViT) [58], Swin Transformer (SwinT) [59], and others, and obtained great performance.

Chen et al. [38] introduced Transformer to the CD task by describing high-level concepts of changes of interest using semantic tokens and modeling collections of tokens to learn their contextual relationships in the spatial and temporal domains rather than directly modeling the dense relationships between all pixels in pixel space. Bandara and Patel [39] developed ChangeFormer, a Transformer-based encoder and decoder that allows the model to capture long-term dependencies at various scales while keeping the long-term dependence information received via the decoder. RSP-BIT [60] employs RS pretraining (RSP) on MillionAID [61] to learn a huge number of BIT-based feature representations for RS images, demonstrating the efficacy of large-scale pretraining in CD tasks. Zhang et al. [62] created SwinSUNet using a hierarchical SwinT structure and a Siamese U-Net, which collects both local and global image information using the local window attention method and provides accurate capture and comprehensive recovery of RS image changes. Ke and Zhang [63] introduced the hybrid Transformer network, which combines CNN and Transformer to address the issues faced by complex sceneries in high-resolution (HR) RS images, particularly for the CD task of irregular and obstructed objects.

Although the aforementioned Transformer-based CD approaches capture the long-range relationships in RS images during training and inference, as well as represent the global semantic content of the images, their downside is that they are computationally intensive.

C. Diffusion- or Frequency-Based RSCD

Diffusion models' excellent image generation capabilities have made them popular in a variety of image processing applications, including image classification [64], object detection [65], image segmentation [66], and CD [67].

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Bandara et al. [67] presented DDPM-CD, which is based on DDPM. The authors employed DDPM for pretraining on the large-scale dataset MillionAID to extract semantic information and then used it as the backbone of the model to aid in the CD tasks. Wen et al. [68] introduced GCD-DDPM, a generative CD model based on differential feature-guided DDPM that uses iterative inference with a variational inference method to dynamically calibrate CD results. Tang and Chen [69] proposed ChangeAnywhere to generate RSCD samples from single-temporal images using the semantic latent diffusion model, which addresses the issue of time-consuming CD sample annotation and the need for expert knowledge by producing large-scale, diverse, and semantically annotated dual-temporal CD datasets. Jia et al. [70] introduced SMDNet to address the weaknesses of CNN and Transformer in capturing boundary features, while also improving the model's resilience under complex lighting and weather situations.

In addition to the diffusion-based RSCD previously discussed, researchers have recently focused their attention on processing RS images in the frequency domain because of its capacity to gain a deeper understanding of the images' semantic information.

To improve the capturing of changes of interest, DDLNet [71] suggests a network architecture based on dual-domain learning (i.e., frequency and spatial domains). Specifically, a frequency-domain enhancement module (FEM) uses the discrete cosine transform (DCT) to extract the frequency components from the input dual time–phase images in order to enhance the changes of interest. To get global interaction features at low computing cost, the global awareness operator (GAO) in GCRNet [72] combines the convolutional theorem with the Fourier transform and the inverse Fourier transform. This method makes indirect use of the frequency-domain data to improve the expressiveness of the model. MDENet [73] introduces the multidomain difference mining network, which aims to address the challenges of image style differences due to seasonal or illumination changes, as well as the similarity of background and foreground features in a scene, and the multidomain difference collaboration module (MDCM), which is capable of accurately capturing the frequencies on different temporal domains and the features of the objects in the spatial domain, and realizes the simultaneous.

Even though the aforementioned frequency-domain processing-based RSCD methods have produced comparatively good detection results, these techniques only process data in the frequency domain; the corresponding network's spatial-domain processing is done independently, and there is no interaction between the frequency-domain and spatial-domain data. This somewhat restricts the network's ability to understand the image's semantic information.

III. Methodology

In this section, we first introduce the knowledge related to the diffusion model; then, present the overall structure of DSFI-CD, and finally explain the individual modules in detail.

A. Preliminaries

Denoising diffusion probabilistic models (DDPMs), an advanced generative model, was first offered as a preliminary idea in 2015 [74], but it was not until 2020 that it was properly described and got extensive attention and use [75]. It has demonstrated outstanding capacity in image generation tasks, producing high-resolution and high-quality images.

The workflow of DDPM for image generation can be divided into two parts, the noise addition process and the denoising process (i.e., sampling). First, in the noise addition process, Gaussian noise is gradually added to the original pre- and post-change image $\mathbf{T}_0 = \{\mathbf{T}_{\text{pre}}, \mathbf{T}_{\text{post}}\} \in \mathcal{R}^{H\times W\times C} \sim p(\mathbf{T}_0)$ until the original image becomes isotropic Gaussian distribution $\mathcal{N}(\mathbf{0}, \boldsymbol{I})$. For the image $\mathbf{T}_t$ at step t, it can be represented by the image $\mathbf{T}_{t-1}$ at step $t-1$ with the following equation:

$$q(\mathbf{T}_t|\mathbf{T}_{t-1}) := \mathcal{N}\left(\mathbf{T}_t; \sqrt{\alpha_t}\mathbf{T}_{t-1}, \sqrt{1-\alpha_t}\boldsymbol{I}\right). \tag{1}$$

Because both the forward noise addition and backward diffusion processes of DDPM can be described by Markov chains, $\mathbf{T}_t$ may be represented by $\mathbf{T}_0$ using the following equation:

$$\mathbf{T}_t = \sqrt{\bar{\alpha}_t}\mathbf{T}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \epsilon \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}) \tag{2}$$

where α_t is the variance scaling factor at time step t. It is usually designed to be monotonically decreasing to ensure a gradual increase in noise. ϵ corresponds to the noise at time step $t = 0$.

During the diffusion process, the diffusion model gradually decreases Gaussian noise to produce a clean image. Whereas the aforementioned DDPM generates different images at random throughout the diffusion process, and while they can increase the number of datasets in the RSCD task, the majority of these generated images are not what we require.

To this goal, some researchers have accomplished controlled image generation by providing some a priori instruction to the model throughout the diffusion process [76], [77], [78]. The difference between conditional and unconditional diffusion is shown in Fig. 1(a) and (b). Referring back to the previously mentioned paper, we begin by characterizing the original image $\mathbf{T}_0$ in the implicit space. Next, we use a basic MLP layer to obtain its embedding coding, which will serve as a prior guide E_0. Finally, we fuse this bootstrap with the original noisy image $\mathbf{T}_t$ to obtain the necessary noisy pseudoimage $\mathbf{T}_{t-1}$. The formula for its calculation is as follows:

$$\begin{aligned} E_0 &= \text{Emb}(\text{MLP}(\varphi(\mathbf{T}_0))) \\ \mathbf{T}_{t-1} &= \text{Fuse}(\mathbf{T}_t, E_0) \end{aligned} \tag{3}$$

where φ denotes the representation operation.

The specific training as well as inference process of conditional diffusion is shown in Algorithm 1.

B. Overall Framwork

Based on the conditional diffusion model's excellent image generation capabilities, we developed DSFI-CD, and the model's overall architecture is shown in Fig. 2. Similar to most existing CD models, we construct DSFI-CD based on encoder–decoder architecture.

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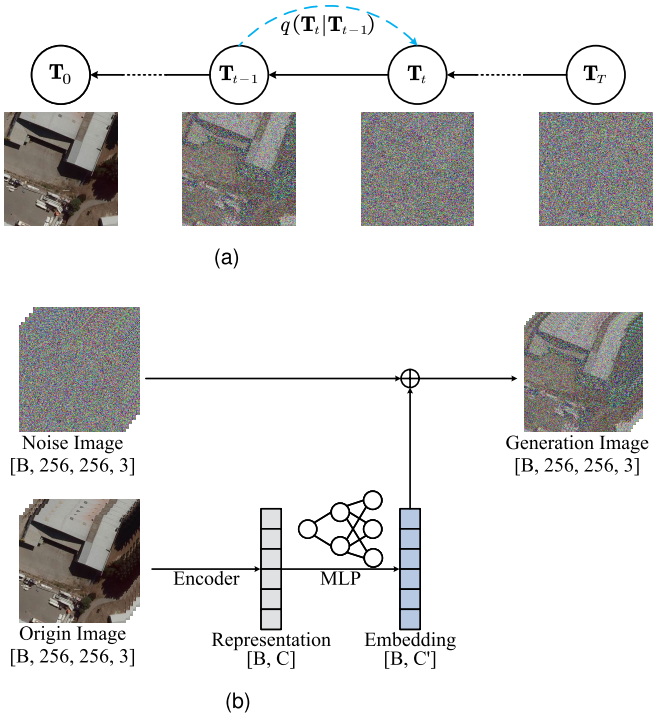


Fig. 1. Schematic of unconditional diffusion and conditional diffusion. (a) Unconditional DDPM. (b) Conditional DDPM.

Algorithm 1 Classifier-Guided Diffusion Sampling, Given a Diffusion Model $(\mu_\theta(\mathbf{T}_t), \Sigma_\theta(\mathbf{T}_t))$, Classifier $p_\phi(E_0|\mathbf{T}_t)$, and Gradient Scale s

Input: class label $\mathbf{T}_0$, gradient scale s
$\mathbf{T}_T \leftarrow$ sample from $\mathcal{N}(0, \mathbf{I})$
for all t from T to 1 **do**
$\mu, \Sigma \leftarrow \mu_\theta(\mathbf{T}_t), \Sigma_\theta(\mathbf{T}_t)$
$\mathbf{T}_{t-1} \leftarrow$ sample from $\mathcal{N}(\mu + s\Sigma\nabla_{\mathbf{T}_t} \log p_\phi(E_0|\mathbf{T}_t), \Sigma$
end for
return $\mathbf{T}_0$

Before feeding the pre- and post-change images $\{\mathbf{T}_{\text{pre}}, \mathbf{T}_{\text{post}}\}$ into the encoder, they are first subjected to a step-by-step noise addition process until they are transformed into two isotropic Gaussian noises, and then, the input image is used as a condition in the diffusion process to conditionally diffuse the aforementioned Gaussian noises, yielding four inputs $\{\mathbf{T}_{\text{pre}}, \mathbf{T}_{\text{post}}, \hat{\mathbf{T}}_{\text{pre}}, \hat{\mathbf{T}}_{\text{post}}\}$ that are fed into the encoder. In this study, we input $\mathbf{T}_{\text{pre}}$ and $\mathbf{T}_{\text{post}}$ into two encoders with shared weights. To better extract the deep semantic features of the input images, we use MiT in SegFormer [79] as the encoder, which has four stages. To produce more representative deep-level features, we combined the $\hat{\mathbf{T}}_{\text{pre}}$ and $\hat{\mathbf{T}}_{\text{post}}$ features extracted from the conditional diffusion process with the features extracted from the original image using MiT. This was accomplished by designing the joint decouple (JD) module to suppress noise from the $\hat{\mathbf{T}}_{\text{pre}}$ and $\hat{\mathbf{T}}_{\text{post}}$ features during the extraction process.

To better decode the deep features acquired from the encoder, we develop two modules in the decoder, SFI module and EE module. In the SFI module, a specific convolutional layer is used to simulate the FFT [80] to extract frequency-domain information in the feature layer, while a series of convolutional layers containing residual modules are used to extract spatial-domain features in the feature layer. The frequency-domain information and spatial-domain features are then interacted to learn the more robust features in the feature layer. In the EE module, we first refer to [56], in which we use the concept of the flow field in the video processing domain to decouple the body of the output features in the SFI module, and then the body features obtained from decoupling and the boundary features are fused in a specific way to improve the boundary features.

The specific training as well as inference process of DSFI-CD is shown in Algorithm 2.

Algorithm 2 Inference of DSFI-CD for CD

Input: $\{\mathbf{T}_{pre}, \mathbf{T}_{post}\} \in \mathcal{R}^{256\times356\times3}$ (A pair of remote sensing images before and after the change)
Output: $\mathbf{M}$ (prediction map)
// step1: Using conditional diffusion model to generate new images while training
$\hat{\mathbf{T}}_{pre} = \text{ConditionDiff}(\mathbf{T}_{pre})$
$\hat{\mathbf{T}}_{post} = \text{ConditionDiff}(\mathbf{T}_{post})$
// step2: In Backbone, the original image is passed through a series of MiT Blocks while the generated image is fused with the main branch using JD Module on the basis of passing through the MiT Blocks.
for n *in* $\{1, 2\}$ **do**
for i *in* $\{1, 2, 3, 4\}$ **do**
$\mathcal{F}_n^i = \text{MiTBlock}(\mathcal{F}_n^{i-1}), \quad where \mathcal{F}_n^0 = \mathbf{T}_{pre}^n$
$\mathscr{F}_n^i = \text{JD}(\text{MiTBlock}(\mathscr{F}_n^{i-1})), \quad where \mathscr{F}_n^0 = \hat{\mathbf{T}_{pre}^n}$
end for
end for
// step3: In Decoder, the two branches in Encoder are spliced in the channel dimension and then input to SFI module for spatial frequency-domain information interaction.
for n *in* $\{1, 2\}$ **do**
$\hat{\mathcal{F}}_n = \text{Concat}(\mathscr{F}_n^4, \mathcal{F}_n^4)$
end for
$\hat{\mathcal{F}}_{out} = \text{SFIModule}(\hat{\mathcal{F}}_1, \hat{\mathcal{F}}_2)$
// step4: After obtaining the output of the SFI Module, it is fed into the EE Module to enhance the boundary information of the feature map.
$\hat{\mathcal{F}}_{body}, \hat{\mathcal{F}}_{edge} = \text{Decouple}(\hat{\mathcal{F}}_{out})$
$\hat{\mathcal{F}}_{pred} = \text{EEModule}(\hat{\mathcal{F}}_{body}, \hat{\mathcal{F}}_{edge}, \mathcal{F}_{rand})$
// step5: Finally a simple classification layer is used to output the final prediction.
$\mathbf{M} = \text{ClsLayer}(\hat{\mathcal{F}}_{pred})$

C. Joint Decouple Module

Comparing the RSCD task's dataset with other RS image interpretation tasks—such as object detection, scene classification, and image segmentation, among others—reveals that it is smaller. Given the rising importance of diffusion models in image synthesis in recent years, we employ a conditional diffusion model for the original input images to indirectly

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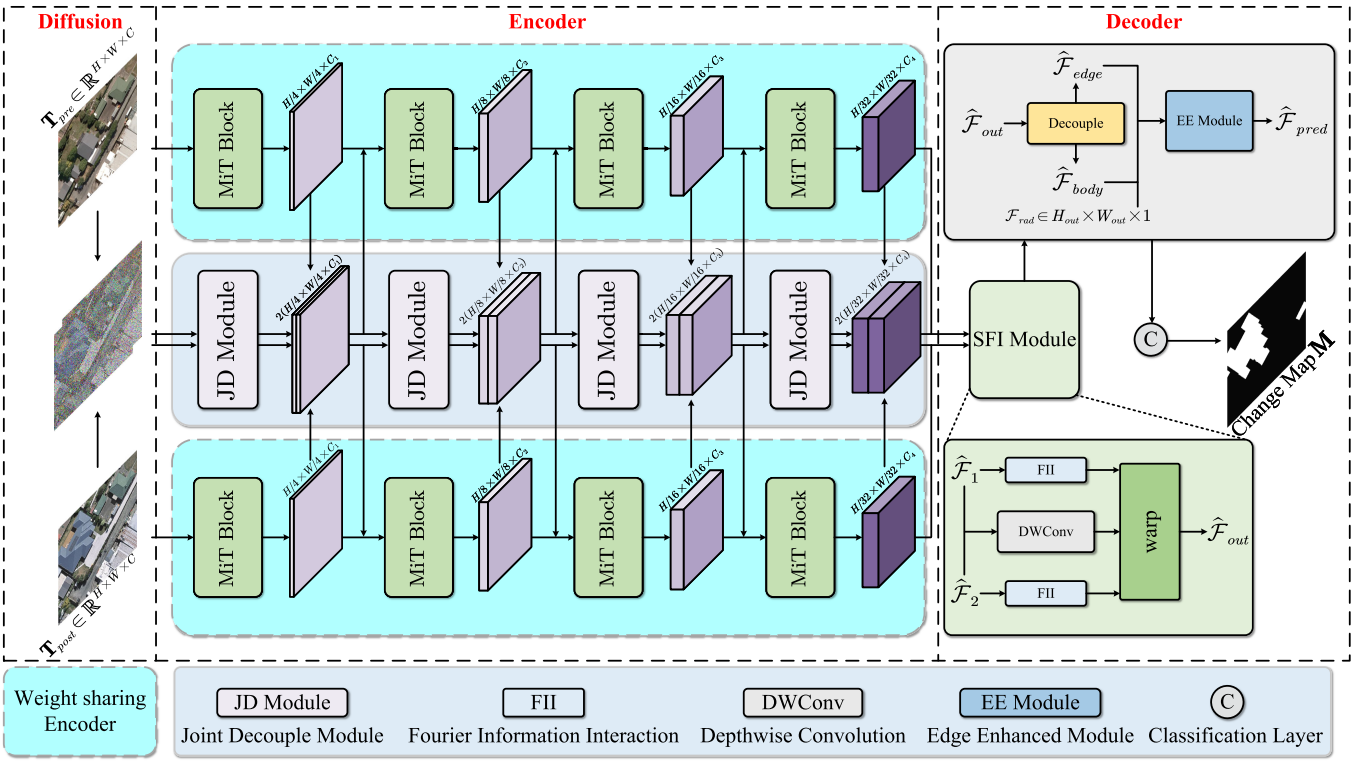


Fig. 2. DSFI-CD overall architectural diagram, which divides the model into three major components: diffusion, encoder, and decoder.

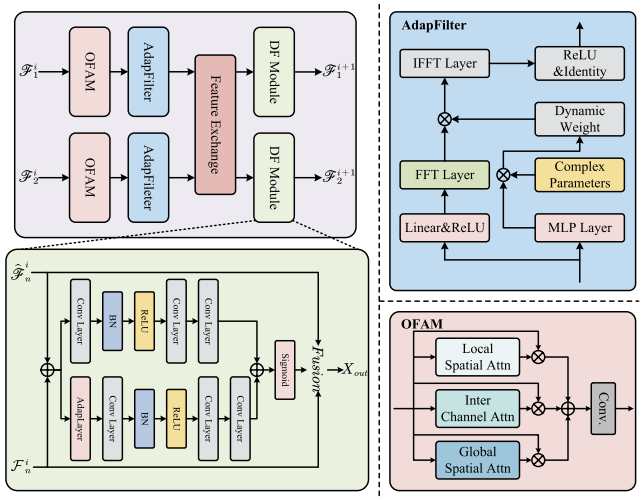


Fig. 3. Schematic of JD the module, with the main structure of the module in the upper left corner, and the DF module, AdapFilter, and OFAM in the lower left, upper right, and lower right corners, respectively.

increase the number of images in the dataset. Furthermore, we created JD module for both the original input images $\{\mathbf{T}_{\text{pre}}, \mathbf{T}_{\text{post}}\} \in \mathcal{R}^{H\times W\times C}$ and the diffusion-generated images $\{\hat{\mathbf{T}}_{\text{pre}}, \hat{\mathbf{T}}_{\text{post}}\} \in \mathcal{R}^{H\times W\times C}$, as shown in Fig. 3. The formula for this step for $\{\hat{\mathbf{T}}_{\text{pre}}, \hat{\mathbf{T}}_{\text{post}}\}$ is as follows:

$$\begin{cases} \mathscr{F}_1^i = \text{MiT}\left(\mathscr{F}_1^{i-1}\right)|_{i=1}^4, \ \text{where } \mathscr{F}_1^0 = \text{MiT}\left(\hat{\mathbf{T}}_{\text{pre}}\right) \\ \mathscr{F}_2^i = \text{MiT}\left(\mathscr{F}_2^{i-1}\right)|_{i=1}^4, \ \text{where } \mathscr{F}_2^0 = \text{MiT}\left(\hat{\mathbf{T}}_{\text{post}}\right) \end{cases} \tag{4}$$

The JD module performs more effective feature extraction on the input first in the local spatial dimension, global spatial dimension, and channel dimension, as demonstrated in the OFAM module in the lower right corner of Fig. 3. $\{\mathscr{F}_1^i, \mathscr{F}_2^i\}|_{i=1}^4$ are first input into MiT, respectively, before being input into the JD module. The calculation formula is as follows:

$$\mathscr{F}_1^{Oi} = \text{ConcatConv.}\begin{pmatrix} \text{LSA}\left(\mathscr{F}_1^i\right) \\ \text{ICA}\left(\mathscr{F}_1^i\right) \\ \text{GSA}\left(\mathscr{F}_1^i\right) \end{pmatrix} \tag{5}$$

where $i = 1, \ldots, 4$, LSA$(\cdot)$, ICA$(\cdot)$, and GSA$(\cdot)$ correspond to local spatial attention, inter channel attention, and global spatial attention in the figure, respectively, and ConcatConv.$(\cdot)$ indicates that the outputs of each are first spliced in the channel dimension, and the spliced results are subsequently convolved by a convolutional kernel size of 3×3.

Then, once the features are extracted in OFAM, we design an adaptive filtering method (AdapFilter) that makes use of FFT and builds complex parameters $\mathcal{A}$ to produce complex vectors that are the same size as the inputs $\{\mathscr{F}_1^{O_i}, \mathscr{F}_2^{O_i}\}|_{i=1}^4$. These complex vectors are then used to create dynamic weights that control the intermediate layer of features produced by the FFT. These weights are then inverted using the inverse FFT (IFFT), the features are mapped back to the original feature domain $\mathcal{D}_o$, and finally, the features are output via a straightforward nonlinear activation method. The formula is expressed as follows:

$$\mathcal{F}_{\text{Filter}}^i = \mathcal{F}^{-1}\begin{pmatrix} \mathcal{F}\left(\zeta\left(\mathscr{F}_1^{O_i}\right)\right) \\ \otimes\psi\left(\mathcal{A}\otimes\left(\xi\left(\mathscr{F}_1^{O_i}\right)\right)\right) \end{pmatrix} \tag{6}$$

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where $i = 1, \ldots, 4$, $\mathcal{F}$ and $\mathcal{F}^{-1}$ stand for the features' respective FFT and IFFT processes, $\mathcal{A}$ indicates the complex parameters in the graph, ζ indicates sending the input features through a linear layer followed by an ReLU operation, ψ indicates the dynamic feature weighting, and ξ indicates passing the input through an MLP layer.

Given that the RSCD task requires consideration of both pre- and post-change features, we use feature exchange in [81] to exchange the pre- and post-change bitemporal features that AdapFilter extracted in the spatial and channel dimensions.

In order to fully fuse the features obtained from the original image by conditional diffusion through the above operation with the features obtained from the original image by MiT, we designed the DF module at the end of the JD module, as shown in the lower left corner of Fig. 3. The DF module uses distinct layers of convolutional modules because the original image deep information branches generated from numerous MiTs and the feature branches from the feature exchange have somewhat different characteristics. To represent the depthwise convolution (DW Conv), we employ numerous Conv Layers, as well as a BN and ReLU in the convolutional module. In another branch, what makes a difference is the use of adaptive mean pooling on top of DW Conv, which is due to the consideration of its good robustness while extracting features efficiently. After extracting the features from the two branches in the DF module, the feature fusion is performed on them. The formula for the DF module is shown in the following equation:

$$\begin{aligned}
\mathcal{F}_{\text{branch1}} &= \sigma\left(\text{Conv}\left(\hat{\mathcal{F}}_n^i \oplus \mathcal{F}_n^i\right)\right) \\
\mathcal{F}_{\text{branch2}} &= \sigma\left(\text{Conv}\left(\text{AdapL}\left(\hat{\mathcal{F}}_n^i \oplus \mathcal{F}_n^i\right)\right)\right) \\
X_{\text{out}} &= 2 \cdot \hat{\mathcal{F}}_n^i \cdot \delta(\mathcal{F}_{\text{branch1}} \oplus \mathcal{F}_{\text{branch2}}) \\
&\quad + 2 \cdot \mathcal{F}_n^i \cdot (1 - \delta(\mathcal{F}_{\text{branch1}} \oplus \mathcal{F}_{\text{branch2}}))
\end{aligned} \tag{7}$$

where AdapL$(\cdot)$ denotes the adaptive mean pooling operation, Conv$(\cdot)$ denotes a convolutional layer with a convolutional kernel size of 3×3, σ denotes batch normalization, ReLU as well as two convolutional layers with a convolutional kernel size of 3×3, and δ denotes the sigmoid operation.

D. SFI Module

To enhance the extraction of semantic information from multiple feature maps via encoder, as well as to combine the information from feature maps in the frequency and spatial domains for additional interaction, we first design the SFI module within the decoder of DSFI-CD, as shown in Fig. 4.

The diagram makes it evident that the SFI module split into two sections: the left half depicts the module's primary organization, while the right half provides a detailed drawing of the FII module. The SFI module is broken into three parts: the primary branch for feature extraction and two FII module branches. In the feature extraction part, the module takes the $\hat{\mathcal{F}}_1$ and $\hat{\mathcal{F}}_2$ fusion through a series of convolutional layers, normalization layers, activation layers, and then further extracts its deep semantic information.

In FII module, after an initial extraction of features by a 3×3 convolutional layer, a series of operations for extracting

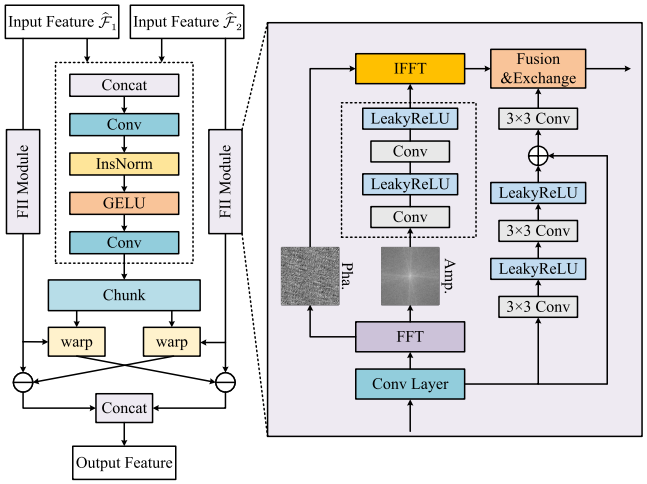


Fig. 4. Schematic of the SFI module, the left side of the figure shows the overall process, and the right side of the figure is one of the FII module.

frequency-domain $\mathcal{D}_{\text{fre}}$ features and for extracting spatial-domain $\mathcal{D}_{\text{spa}}$ features are processed in parallel.

For the frequency domain $\mathcal{D}_{\text{fre}}$, the feature map's time-domain signal is first transformed to a frequency-domain representation using the FFT, which decomposes the signal into frequency component combinations. In this step, the FFT retrieves 2 bits of information: amplitude $\mathcal{I}_{\text{amp}}$ and phase $\mathcal{I}_{\text{pha}}$. The amplitude information depicts the strength of each frequency component of the signal, whereas the phase information provides the temporal position or relative time offset of each frequency component.

Furthermore, because the amplitude information is closely tied to signal intensity, increasing or decreasing amplitude can effectively emphasize the signal's main qualities while lowering or eliminating the influence of noise. To extract semantic information from the amplitude component, the FII module employs two 3×3 convolutional layers and two activation layers. Because the phase component is more difficult to process and phase misalignment might cause phase distortion, we do not process it further and instead feed it straight into the IFFT alongside the treated amplitude component. Furthermore, in the spatial-domain dimension, we utilize a combination of 3×3 convolutional layers with residual structures and activation layers. The computational procedure in the frequency and spatial domains is shown in the following equation:

$$\mathcal{F}_F = \mathcal{F}^{-1}\left(\underbrace{\mathcal{F}\left(\text{Conv}\left(\hat{\mathcal{F}}_1\right)\right)}_{\text{phase component } \mathcal{I}_{\text{pha}}}, \zeta \quad \underbrace{\mathcal{F}\left(\text{Conv}\left(\hat{\mathcal{F}}_1\right)\right)}_{\text{amplitude component } \mathcal{I}_{\text{amp}}}\right) \tag{8}$$

$$\mathcal{F}_S = \text{Conv}\left(\zeta\left(\text{Conv}\left(\hat{\mathcal{F}}_1\right)\right) \oplus \text{Conv}\left(\hat{\mathcal{F}}_1\right)\right) \tag{9}$$

where Conv denotes a convolutional layer with a convolutional kernel size of 3×3, $\mathcal{F}$ denotes FFT, ζ denotes the two consecutive convolutional layers with a convolutional kernel size of 3×3 as well as LeakyReLU, $\mathcal{F}^{-1}$ denotes IFFT, and $\oplus$ denotes the concat operation in the channel dimension.

Following the extraction of features in the frequency and spatial domains, the feature interaction operation—referred to

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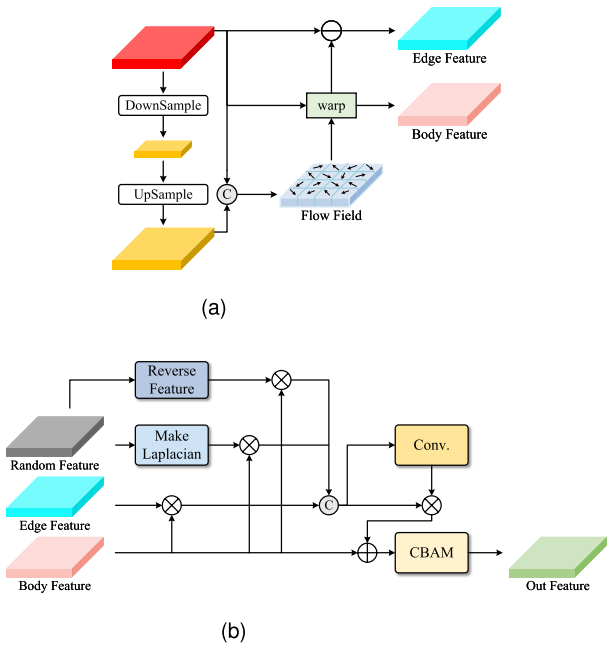


Fig. 5. Schematic of EE module. (a) Decouple features. (b) Edge enhance.

as fusion and exchange in Fig. 4—as well as the fusion of the two are carried out. Here, we simultaneously input the spatial-domain information extracted in the previous operation into the frequency-domain information extraction operation and the frequency-domain information extracted from the previous operation into the spatial-domain information extraction operation. This step is shown in the following equation:

$$\hat{\mathcal{F}}_F = \mathcal{F}^{-1}\left(\underbrace{\mathcal{F}(\mathrm{Conv}(\mathcal{F}_S))}_{\text{phase component } \mathcal{I}_{\text{pha}}}, \zeta \quad \underbrace{\mathcal{F}(\mathrm{Conv}(\mathcal{F}_S))}_{\text{amplitude component } \mathcal{I}_{\text{amp}}}\right) \tag{10}$$

$$\hat{\mathcal{F}}_S = \mathrm{Conv}(\zeta(\mathrm{Conv}(\mathcal{F}_F)) \oplus \mathrm{Conv}(\mathcal{F}_F)) \tag{11}$$

where $\hat{\mathcal{F}}_F$ and $\hat{\mathcal{F}}_S$ are the frequency-domain and spatial-domain features obtained after the fusion and exchange operation, respectively.

Subsequently, after obtaining $\hat{\mathcal{F}}_F$ and $\hat{\mathcal{F}}_S$, the two are fused with the aforementioned deep information extracted from $\hat{\mathcal{F}}_1$ and $\hat{\mathcal{F}}_2$ through a series of convolutional layers to output the result of SFI module $\hat{\mathcal{F}}_{\text{out}}$

$$\mathcal{F}^n_{\text{out}} = \mathrm{Concat}\left(\hat{\mathcal{F}}^n_F, \hat{\mathcal{F}}^n_S\right), \text{where } n \in \{1, 2\} \tag{12}$$

$$\hat{\mathcal{F}}_{\text{out}} = \mathrm{Concat}\begin{pmatrix} \mathrm{warp}_1\left(\Psi\left(\hat{\mathcal{F}}_1, \hat{\mathcal{F}}_2\right), \mathcal{F}^1_{\text{out}}\right) \Theta \mathcal{F}^2_{\text{out}}, \\ \mathrm{warp}_2\left(\Psi\left(\hat{\mathcal{F}}_1, \hat{\mathcal{F}}_2\right), \mathcal{F}^2_{\text{out}}\right) \Theta \mathcal{F}^1_{\text{out}} \end{pmatrix} \tag{13}$$

where Ψ denotes a series of convolutional layers, warp reference to the flow field characteristics of its ideas [82], the input feature layer is corrected to prevent the emergence of the feature map of the points in the phenomenon of offset, and Θ denotes the front and back of the two are subtracted.

E. EE Module

To further increase the model's detection accuracy for the boundary of the changing region in decoder, we built the EE module, as shown in Fig. 5(a) and (b).

1) Decouple Features: In Fig. 5(a), the input $\hat{\mathcal{F}}_{\text{out}}$ is first down- and up-sampled sequentially in order to learn more semantic information while maintaining the same feature map size. Then, the input $\hat{\mathcal{F}}_{\text{out}}$ and the sampled result $\mathrm{DownUp}(\hat{\mathcal{F}}_{\text{out}})$ are spliced in the channel dimensions through a 3×3 convolutional layers, we make the result obtained from the above operation the flow field.

Once the flow field is obtained, we input the flow field and $\hat{\mathcal{F}}_{\text{out}}$ into the warp. The bilinear interpolation is then used to align the points in the flow field using the input, avoiding the effect of the previously described upsampling and downsampling operation, which causes the edges of the changing regions in the feature map to become offset. In this case, the computational flow of warp is shown as follows:

$$\mathrm{Warp}(\cdot) \Leftarrow \begin{cases} \Delta_{l-1} = \mathrm{Conv}_l(\mathrm{Concat}(\mathcal{F}_l, \mathcal{F}_{l-1})) \\ p_l = p_{l-1} + \dfrac{\Delta_{l-1}(p_{l-1})}{2} \\ \mathcal{F}_l(p_{l-1}) = \mathcal{F}_l(p_l) = \displaystyle\sum_{p \in N(p_l)} \omega_p \mathcal{F}_l(p) \end{cases} \tag{14}$$

we do not strictly specify the formula in (14) based on the meaning of each symbol used in this article because the idea of warp is applicable to other visual tasks in addition to the CD task. Where $\mathcal{F}_l$ and $\mathcal{F}_{l-1}$ denote the input feature maps at the correct location as well as the feature maps with correction, respectively, Conv_l denotes a simple 3×3 convolutional layer, $N(p_l)$ denotes the neighborhood of the deformed point p_l in $\mathcal{F}_l$, and ω_p denotes the bilinear kernel weights.

The input $\hat{\mathcal{F}}_{\text{out}}$ can be separated into more robust body feature $\mathcal{F}_{\text{body}}$ and edge feature $\mathcal{F}_{\text{edge}}$ by using warp. The relevant formulae are as follows:

$$\begin{cases} \mathcal{F}_{\text{flow}} &= \mathrm{ConcatConv}(\mathcal{F}_{in}, \mathrm{DownUp}(\mathcal{F}_{\text{in}})) \\ \mathcal{F}_{\text{body}} &= \mathrm{Warp}(\mathcal{F}_{\text{flow}}, \mathcal{F}_{\text{in}}) \\ \mathcal{F}_{\text{edge}} &= \mathcal{F}_{\text{in}} - \mathcal{F}_{\text{body}} \end{cases} \tag{15}$$

2) Edge Enhance: To increase the accuracy of the change boundary, we believe that decoupling the boundary and the feature map's body alone is insufficient, and that additional processing is necessary after decoupling the features. As shown in Fig. 5(b), we perform a series of operations on the edge and body features generated above to enhance the boundary information.

The general concept of the module is depicted in the figure as multifeature fusion, i.e., fusing the edge feature $\mathcal{F}_{\text{edge}}$ with the body feature $\mathcal{F}_{\text{body}}$ acquired in decouple feature using a novel technique. It is noteworthy that we present random feature $\mathcal{F}_{\text{rand}}$, taking advantage of Laplace pyramid's superior performance in the boundary extraction [83], we apply it to random feature $\mathcal{F}_{\text{rand}}$ and combine it with body feature $\mathcal{F}_{\text{body}}$ in order to improve the change region's boundary information. The computational flow of the module is shown as follows:

$$\mathcal{F}_{\text{conv}} = \mathrm{ConcatConv}\begin{pmatrix} \left(\mathcal{F}_{\text{body}} \otimes \mathcal{F}_{\text{edge}}\right) \\ \left(\mathcal{F}_{\text{body}} \otimes \mathcal{F}_{\text{rand}}\right) \\ \left(\mathcal{F}_{\text{body}} \otimes (1 - \mathcal{F}_{\text{rand}})\right) \end{pmatrix} \tag{16}$$

where ConcatConv denotes that the feature map is first spliced in channel dimension and subsequently passed through a 3×3 convolutional layer and $\otimes$ denotes the pixel-by-pixel

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multiplication of the two feature maps. We simply compute the background information of the input feature map using $1 - \mathcal{F}_{\text{rand}}$, such that when extracting information from the foreground (change region), the background information is taken into consideration.

The output $\mathcal{F}_{\text{out}}$ of the EE module is represented by the following equation at the end, with reference to the CBAM attention mechanism [84]. Also, the final result $\mathbf{M}$ will be output by $\hat{\mathcal{F}}_{\text{pred}}$ through a simple classification layer

$$\hat{\mathcal{F}}_{\text{pred}} = \text{CBAM}\left(\mathcal{F}_{\text{body}} + \mathcal{F}_{\text{conv}} \otimes \text{Conv}(\mathcal{F}_{\text{conv}})\right) \tag{17}$$

$$\mathbf{M} = \text{ClsLayer}\left(\hat{\mathcal{F}}_{\text{pred}}\right) \tag{18}$$

IV. Experiments

This section introduces the datasets that were used for the experiments in the article, describes the experimental setup, compares the models from previous years, analyzes the experimental results of DSFI-CD on various datasets, and conducts ablation experiments to verify the significance of the model's components, which are illustrated using intermediate layer feature maps.

A. Datasets

Experiments were carried out on the following four datasets in order to completely validate the efficacy of DSFI-CD.

1) WHU-CD [85]: The WHU-CD CD dataset, a publicly available dataset dedicated to building CD, was obtained by Wuhan University researchers from aerial imageries collection from Christchurch, New Zealand, which collected pre- and post-change imagery from 2012 to 2016, with a spatial resolution of 0.3 m/pixel. The dataset spans an area of 20.5 km^2 and includes 12 796 buildings in 2012 and 16 077 buildings in 2016, reflecting urban redevelopment in the city prior to and following the earthquake. In the experiments presented in this study, we crop the images before and after the change to a size of 256 × 256 for convenience, and the number of images in the training set, validation set, and test set are 5947, 744, and 743, respectively.

2) CLCD [86]: The Cropland CD (CLCD) dataset is a high-resolution dataset specifically designed for CLCD. It consists of 600 pairs of bitemporal images collected by the Gaofen-2 satellite in Guangdong, China, in 2017 and 2019. The images have a spatial resolution ranging from 0.5 to 2 m/pixel. Each pair of images is of size 512 × 512 pixels and is accompanied by a binary label indicating cropland changes. The dataset includes various types of changes such as buildings, roads, lakes, and bare soil lands. For experimental purposes, the dataset is divided into 320 pairs for training, 120 pairs for validation, and 120 pairs for testing.

3) SYSU-CD [53]: The dataset is a publicly available, multichange-category yet change-category-independent dataset of aerial images from Hong Kong collected by Sun Yat-sen University researchers between 2007 and 2014. The dataset includes a wide range of change categories (e.g., urban buildup, suburban sprawl, infrastructure, vegetation change, road expansion, and marine buildup), with a particular emphasis on metropolitan and coastal areas of the city. The dataset contains 20 000 256 × 256 before and after changing image pairs, and we configured the training, validation, and test sets at a 6:2:2 ratio to facilitate experiments.

4) CDD [87]: The CDD dataset is an open dataset captured from Google Earth and contains 11 raw image pairs, each with a resolution of 0.03–1 m/pixel, of which seven pairs are seasonal change images (4.725 × 2.700) and four pairs are disaster change images (1.900 × 1.000), covering different geographic regions and landscape types, and each image pair contains complex changing scenarios, such as cloud cover, shadows, lighting, and rotation. The majority of the collection contains seasonal variations in natural objects (e.g., from single trees to enormous wooded regions), with a minor component containing changes in items of various sizes (e.g., vehicles to large architectural constructions). In this study, to simplify model training, we crop each original image into a size of 256 × 256 images, resulting in a dataset of 16 000 images, including 10 000/3000/3000 images for training, validation, and testing sets.

B. Experimental Setup

1) Training Details: In the experiments of this article, we have adopted the Python 3.10.4 programming language, based on the PyTorch 2.0.1 DL framework [88], and used the MMSegmentation-based Opencd toolkit [81] as a training as well as testing environment. Hardwarewise, we employed an NVIDIA RTX 4090 GPU, 32 GB of RAM, and an Intel Xeon E5-2678 v3 at 2.50 GHz×2. Furthermore, for hyperparameters, we employed AdamW [89], which is officially defined as follows, as the optimizer and BCE loss, the loss function described in the research, for our trials. The BCEloss formula is calculated as follows:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{H \times W} \sum_{h=1,w=1}^{H,W} \left[\mathbf{Y}(h,w) + (1-\mathbf{Y}(h,w)) \cdot \log\left(1-\hat{\mathbf{Y}}(h,w)\right)\right] \tag{19}$$

where H and W denote the height and width of each image, respectively, $\mathbf{Y}(h, w)$ is the predicted value of the (h, w) point in each image, and $\hat{\mathbf{Y}}(h, w)$ is the true value of the (h, w) point in each image.

2) Evaluation Metrics: The performance of each model is evaluated using the following evaluation metrics in this study so that the models provided in this research can be properly compared with other models.

1) *Precision (Pre):* Indicates the ratio of the number of correctly identified change region pixels to all correctly identified pixels. Expressed by the following equation:

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{20}$$

2) *Recall (Rec):* Indicates the ratio of the number of non-change region pixels that were correctly identified to all the pixels that were correctly identified. Expressed by the following equation:

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{21}$$

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3) *F1 Score (F1):* This indicator is used to synthesize the rating of Pre and Rec. Expressed by the following equation:

$$F1 = 2 \times \frac{\text{Pre} \times \text{Rec}}{\text{Pre} + \text{Rec}} \tag{22}$$

4) *Intersection over Union (IoU):* Indicates the ratio of the regions in the prediction map that is the same as in the labeling map to the whole image. Expressed by the following equation:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{23}$$

5) *Overall Acccuracy (OA):* Indicates the proportion of correctly categorized pixels to the total number of pixels. Expressed by the following equation:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{24}$$

6) *Cohen's Kappa ($\mathcal{K}_c$):* A statistic to assess the level of agreement between two classifiers in categorizing data. Expressed by the following equation:

$$\mathcal{K}_c = \frac{p_o - p_e}{1 - p_e} \tag{25}$$

In (20)–(24), TP denotes the number of positive samples (i.e., change region pixels) that were correctly identified, FP denotes the number of pixels that were incorrectly identified as positive samples, and FN denotes the number of pixels that were incorrectly identified as negative samples (i.e., nonchange regions). In (25), p_o is the ratio of actual consistency and p_e is the expected ratio of chance consistency.

C. Comparison With SOTA Methods

To completely demonstrate the superiority of DSFI-CD in the RSCD task, we used the following SOTA models from recent years to conduct tests on the four datasets described above.

1) *FC-EF, FC-Siam-Di, and FC-Siam-Conc* [33] using full convolutional networks [31], constructs three models for the first time that are similar to U-Net networks [32] for RSCD tasks. These models use various approaches to fuse dual-temporal RS imageries.
2) *BIT* [38] first applied Transformer [37] to the RSCD task, by employing the Transformer design to effectively capture long-distance conceptual connections in the spatiotemporal domain and model contextual information in the spatiotemporal domain.
3) *ChangeFormer* [39] differs from the traditional use of CNN to extract features from the input image by combining a hierarchical Transformer encoder and an MLP decoder in a Siamese network framework to efficiently render long-range details at multiple scales.
4) *ChangerEx-MiT* [81] simultaneously takes into account the information before and after changes in remotely sensed imagery and proposes different feature exchange strategies—AD and feature "exchange."
5) *HANet* [54] addresses the pixel imbalance between changed and unchanged regions in the RSCD task by proposing a stepwise foreground-balanced sampling strategy to improve the model learning for changed pixels, as well as employing an approach with a hierarchical focus to integrate multiscale features for finer detection.
6) *SNUNet* [90], with close hopping connections between the encoder and decoder as well as inside the decoder, SNUNet is able to preserve high-resolution fine-grained features while lowering pixel uncertainty at shifting target boundaries and deterministic omission of small targets.
7) *STANet* [91] captures spatial–temporal correlations in RSCD task through a self-attention mechanism to generate more discriminative features. It is organized into three variants: STANet-Base, STANet-Bam, and STANet-Pam.
8) *TINY-CD* [46] optimizes the exploitation of low-level information in the spatial and temporal domains by utilizing the Siamese U-Net architecture and a novel feature mixing approach. It also offers a unique mechanism for spatial semantic attention through its MAMB.
9) *MAFGNet* [92] makes use of the tremendous relational learning capability of graphical convolutional neural networks to investigate long-term associations between the target detection region and the global at the spatial and channel levels, as well as to successfully fuse multiscale feature information.
10) *ELGC-Net* [93], the efficient local–global context aggregator (ELGCA) module is the main component of ELGC-Net. It is integrated into the encoder and captures enhanced global context and local spatial information using a revolutionary pooled-transpose (PT) attention method and deep convolution.
11) *LightCDNet* [94] is a lightweight CD network designed to maintain high performance while drastically reducing model size for industrial applications and edge devices.
12) *BAN* [95] provides a novel learning paradigm, a broader framework for RSCD based on the underlying model. The goal of BAN is to extract knowledge from the underlying model to help with CD and lessen the dependency on vast amounts of labeled data.
13) *STNet* [57] is built with two feature fusion mechanisms: a spatial feature fusion module that recovers the spatial details of the change representations by capturing the fine-grained geographic information through a cross-scale attention mechanism, and a TFF module that emphasizes and combines the change points of interest in dual-temporal images through a cross-temporal gating mechanism.
14) *DDLNet* [71] is an RSCD network based on the dual-domain learning (frequency and spatial domains). The fundamental innovation of DDLNet is the construction of FEM and spatial-domain recovery module (SRM) to increase the performance of CD.
15) *DDPM-CD* [67] uses DDPMs as a feature extractor to detect changes. Unconditional DDPM is initially pretrained on a large number of unlabeled RS images in order to understand the important semantics. This

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TABLE I

EXPERIMENTAL RESULTS OF DSFI-CD WITH OTHER SOTA METHODS ON BOTH DSIFN-CD AND CDD DATASETS, WHERE RED INDICATES THAT THE METRIC REACHES THE BEST AMONG ALL METHODS, FOLLOWED BY BLUE, AND BLACK IS THE THIRD HIGHEST METRIC, ALL INDICATORS ARE EXPRESSED IN PERCENTAGES (%). THE NUMBER LABELED IN THE LOWER RIGHT CORNER OF EACH METHOD NAME IS THE YEAR THE METHOD WAS PROPOSED. IT IS WORTH NOTING THAT "-" IN THE BACKBONE COLUMN INDICATES THAT THE METHOD DOES NOT USE A BACKBONE FROM A WIDELY USED CLASSIFICATION MODEL, BUT INSTEAD CONSTRUCTS A SPECIFIC BACKBONE

	Methods	Backbone	CLCD						CDD					
			F1	Pre	Rec	IoU	$\mathcal{K}_c$	OA	F1	Pre	Rec	IoU	$\mathcal{K}_c$	OA
SOTA methods	FC-EF$_{18}$ [33]	UNet	65.55	66.11	65.0	48.75	62.8	94.92	66.86	82.17	56.36	50.21	63.29	93.33
	FC-Siam-Di$_{18}$ [33]	UNet	66.85	69.88	64.07	50.21	64.31	95.27	73.86	91.98	61.71	58.56	71.09	94.79
	FC-Siam-Conc$_{18}$ [33]	UNet	66.03	65.28	66.8	49.29	63.27	94.89	77.19	90.29	67.41	62.85	74.6	95.25
	BIT$_{22}$ [38]	ResNet18	66.62	76.83	58.81	49.95	64.32	95.62	94.25	95.44	93.09	89.13	93.48	98.64
	ChangeFormer$_{22}$ [39]	SegformerB0	73.07	76.44	69.99	57.57	71.01	96.16	86.42	94.86	79.36	76.09	84.79	97.05
	ChangerEx-MiT$_{23}$ [81]	SegformerB0	**77.13**	**80.65**	**73.91**	**62.78**	**75.38**	**96.74**	95.65	96.31	95.0	91.66	95.06	98.97
	HANet$_{23}$ [54]	-	62.16	67.25	57.79	45.1	59.37	94.77	96.19	96.94	95.44	92.65	95.67	99.1
	SNUNet$_{22}$ [90]	NestedUNet	60.01	56.36	64.17	42.87	56.57	93.64	**96.59**	**97.35**	95.84	**93.4**	**96.13**	**99.19**
	STANet-Base$_{20}$ [91]	ResNet18	56.56	54.03	59.34	39.43	52.89	93.22	93.76	91.52	96.1	88.25	92.89	98.47
	STANet-Bam$_{20}$ [91]	ResNet18	46.18	76.27	33.12	30.02	43.64	94.26	93.12	89.76	**96.73**	87.12	92.14	98.29
	STANet-Pam$_{20}$ [91]	ResNet18	56.11	56.7	55.53	38.99	52.62	93.54	95.24	93.45	**97.09**	90.91	94.58	98.84
	TINY-CD$_{23}$ [46]	EfficientnetB4	67.32	74.35	61.5	50.73	64.95	95.56	95.57	96.24	94.91	91.51	94.97	98.95
	MAFGNet$_{24}$ [92]	-	57.57	63.84	52.42	40.42	54.52	94.25	91.79	93.17	90.45	84.82	90.71	98.09
	ELGC-Net$_{24}$ [93]	-	58.86	59.22	58.5	41.7	55.57	93.91	92.32	93.7	90.98	85.74	91.31	98.22
	LightCDNet-S$_{23}$ [94]	-	53.61	61.51	47.51	36.62	50.39	93.88	92.61	94.14	91.13	86.24	91.63	98.26
	LightCDNet-B$_{23}$ [94]	-	63.12	66.05	60.43	46.11	60.29	94.74	95.62	96.51	94.75	91.61	95.03	98.96
	LightCDNet-L$_{23}$ [94]	-	60.99	71.08	53.4	43.87	58.33	94.92	96.05	96.46	95.64	92.39	95.51	99.06
	STNet$_{23}$ [57]	ResNet18	74.99	76.91	73.17	59.99	73.04	96.37	95.7	98.13	93.39	91.75	95.13	99.0
	DDLNet$_{24}$ [71]	ResNet18	75.03	75.92	**74.16**	60.03	73.04	96.33	95.35	98.16	92.69	91.11	94.74	98.92
	BAN$_{24}$ [95]	ViT-B	66.19	65.38	67.02	49.46	63.43	94.91	94.44	95.23	93.66	89.46	93.69	98.68
	DDPM-CD$_{22}$ [67]	-	71.78	74.3	69.43	55.99	69.6	95.94	91.55	93.89	89.33	84.42	90.51	98.15
	MambaBCD-Base$_{24}$ [45]	-	**76.61**	**79.49**	**73.94**	**62.09**	**74.81**	**96.64**	**96.55**	**98.01**	95.12	**93.32**	**96.09**	**99.19**
	DSFI-CD	SegformerB0	**78.94**	**84.74**	73.88	**65.2**	**77.37**	**97.07**	**98.01**	**97.68**	**98.34**	**96.1**	**97.74**	**99.52**

pretrained DDPM is then utilized to extract features for RSCD.

16) *MambaBCD* [45] is a BCD network based on the Mamba architecture, designed to detect surface changes using multitemporal RS images. It employs the visual state-space model (VMamba) as the encoder, which can fully learn the global contextual information of input images with linear complexity.

In fairness, the methods described above in this article make use of the identical hyperparameter settings and hardware setup as DSFI-CD in Section IV-B1.

D. Quantitative Analysis

The evaluation metrics of all the models used in this article's experiments on the test sets of the four datasets—CLCD, CDD, WHU-CD, and SYSU-CD—are presented quantitatively in Tables I and II. To highlight the methods with greater detection performance in the table, the best, second best, and third best values in each evaluation metric are given in red, blue, and black. It is apparent that DSFI-CD outperforms all other models in terms of most evaluation metrics in each dataset. Specifically, on the CLCD dataset, DSFI-CD improves the measure of IoU by 2.42% and 3.11% compared wiuth ChangerEx-MiT, which also utilizes SegFormerB0 as a backbone and MambaBCD-Base, and for DDPMs, which also introduce DDPMs for CD task, the IoU is only 55.99%.

Although many methods perform better on the relatively easy-to-detect CDD dataset, DSFI-CD still shows some improvement over other methods. For example, in terms of IoU metrics, DSFI-CD reaches 96.1%, the only method to surpass 95%, and improves by 2.7% over the suboptimal method. These results show that DSFI-CD performs well on the detection results on the CDD dataset in some details, such as edges, small targets, and so forth.

Similar to the results on CLCD and CDD, on the two datasets WHU-CD and SYSU-CD, DSFI-CD outperforms other methods in six evaluation metrics, and the model in this article improves the $F1$ and IoU on these two datasets compared with other models compared with the suboptimal model by 0.35%, 0.64%, 1.89%, and 2.86%. WHU-CD is a building CD dataset with favorable illumination circumstances, thus we believe that what distinguishes DSFI-CD from other approaches is the high performance of the EE module in DSFI-CD for edge detection.

Together with the previously mentioned outcomes, we additionally calculated how many parameters (Params) and floating points of computation (FLOPs) each method required. The results are displayed in Table III, which demonstrates that while DSFI-CD's Params and FLOPs are not the smallest, they are still sufficiently small in comparison with the majority of the methods.

E. Quantitative Visualization

To demonstrate DSFI-CD's superiority over various methods on different datasets, we randomly chose some representative images from the test set from several datasets for testing, and the results are displayed in Figs. 6–9. Within each illustration, correctly recognized regions in the foreground (changed) and background (unchanged) are shown in white and black, respectively, whereas incorrectly detected and undetected regions are shown in red and cyan.

1) Visualization on CLCD Dataset (Fig. 6): The main objective of the CLCD dataset is cropland CD. It is worth mentioning that CLCD is compared with other datasets, in which

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TABLE II

EXPERIMENTAL RESULTS OF DSFI-CD WITH OTHER SOTA METHODS ON BOTH WHU-CD AND SYSU-CD DATASETS, WHERE RED INDICATES THAT THE METRIC REACHES THE BEST AMONG ALL METHODS, FOLLOWED BY BLUE, AND BLACK IS THE THIRD HIGHEST METRIC, ALL INDICATORS ARE EXPRESSED IN PERCENTAGES (%). THE NUMBER LABELED IN THE LOWER RIGHT CORNER OF EACH METHOD NAME IS THE YEAR THE METHOD WAS PROPOSED. IT IS WORTH NOTING THAT "-" IN THE BACKBONE COLUMN INDICATES THAT THE METHOD DOES NOT USE A BACKBONE FROM A WIDELY USED CLASSIFICATION MODEL, BUT INSTEAD CONSTRUCTS A SPECIFIC BACKBONE

	Methods	Backbone	WHU-CD						SYSU-CD					
			F1↑	Pre↑	Rec↑	IoU↑	$\mathcal{K}_c$↑	OA↑	F1↑	Pre	Rec	IoU↑	$\mathcal{K}_c$↑	OA↑
SOTA methods	FC-EF$_{18}$ [33]	UNet	80.64	81.98	79.34	67.56	79.65	98.12	76.05	76.39	75.7	61.35	69.14	89.28
	FC-Siam-Di$_{18}$ [33]	UNet	67.64	54.69	88.61	51.1	65.54	95.82	66.97	**91.25**	52.89	50.34	60.44	88.27
	FC-Siam-Conc$_{18}$ [33]	UNet	64.66	51.47	86.94	47.78	62.33	95.32	76.25	84.36	69.57	61.62	70.19	90.26
	BIT$_{22}$ [38]	ResNet18	90.97	90.78	91.17	83.44	90.5	99.11	62.7	47.09	**93.81**	45.67	46.76	74.91
	ChangeFormer$_{22}$ [39]	SegformerB0	92.07	94.48	89.78	85.3	91.67	99.24	83.35	**89.5**	77.99	71.45	78.93	92.99
	ChangerEx-MiT$_{23}$ [81]	SegformerB0	**93.35**	94.08	92.62	**87.53**	**93.01**	**99.35**	83.35	87.47	79.59	71.45	78.8	92.85
	HANet$_{23}$ [54]	-	89.73	90.2	89.26	81.37	89.2	98.99	79.5	82.81	76.45	65.98	73.86	91.14
	SNUNet$_{22}$ [90]	NestedUNet	90.29	90.83	89.75	82.3	89.79	99.05	79.87	85.6	74.85	66.48	74.52	91.52
	STANet-Base$_{20}$ [91]	ResNet18	85.36	79.0	**92.84**	74.46	84.54	98.43	74.12	80.22	68.88	58.88	67.33	89.18
	STANet-Bam$_{20}$ [91]	ResNet18	87.12	82.85	91.85	77.18	86.42	98.66	76.74	75.49	78.03	62.26	69.85	89.37
	STANet-Pam$_{20}$ [91]	ResNet18	88.07	85.97	90.28	78.68	87.44	98.79	79.02	78.49	79.56	65.32	72.49	90.04
	TINY-CD$_{23}$ [46]	EfficientnetB4	92.35	93.15	91.56	85.79	91.96	99.25	80.68	79.85	**81.53**	67.62	75.0	91.22
	MAFGNet$_{24}$ [92]	-	91.86	**94.53**	89.35	84.95	91.54	99.21	81.54	83.65	79.53	68.84	76.36	91.9
	ELGC-Net$_{24}$ [93]	-	87.54	87.49	87.59	77.85	87.03	98.84	79.11	82.86	75.69	65.44	73.4	91.01
	LightCDNet-S$_{23}$ [94]	-	90.81	90.57	91.05	83.17	90.33	99.09	82.71	86.56	79.19	70.52	77.98	92.56
	LightCDNet-B$_{23}$ [94]	-	93.08	94.52	91.68	87.06	92.73	99.33	83.14	85.54	80.87	71.15	78.43	92.63
	LightCDNet-L$_{23}$ [94]	-	91.74	92.22	91.25	84.73	91.31	99.19	82.62	86.12	79.39	70.38	77.83	92.49
	STNet$_{23}$ [57]	ResNet18	91.0	90.91	91.1	83.49	90.54	99.11	83.28	87.97	79.06	71.35	78.76	92.86
	DDLNet$_{24}$ [71]	ResNet18	88.8	87.08	90.59	79.86	88.21	98.87	83.61	**90.16**	77.95	71.83	79.29	**93.13**
	BAN$_{24}$ [95]	ViT-B	89.34	92.98	85.97	80.73	88.81	98.99	78.25	84.31	72.99	64.26	72.51	90.87
	DDPM-CD$_{22}$ [67]	-	89.47	88.66	90.28	80.94	88.91	98.95	**83.94**	87.88	80.34	**72.32**	**79.55**	93.09
	MambaBCD-Base$_{24}$ [45]	-	**94.15**	**95.33**	**92.99**	**88.94**	**93.85**	**99.43**	**84.05**	89.01	79.61	**72.49**	**79.75**	**93.21**
	DSFI-CD	SegformerB0	**94.5**	**95.82**	**93.22**	**89.58**	**94.22**	**99.47**	**85.94**	89.23	**82.89**	**75.35**	**82.06**	**93.9**

TABLE III

NUMBER OF PARAMETERS (PARAMS) FOR EACH MODEL AND THE COMPARISON OF FLOATING POINT NUMBERS (GFLOPS) WERE DONE USING AN INPUT IMAGE SIZE OF 256 × 256 × 3

Methods	Params(M)	GFLOPs(G)	Avg Time(s)
FC-EF	1.35	12.98	21
FC-Siam-Di	1.35	17.54	15
FC-Siam-Conc	1.55	19.96	16
BIT	2.99	34.99	9
ChangeFormer	3.85	11.38	23
ChangerEx-MiT	3.46	8.52	16
HANet	3.03	97.55	43
SNUNet	3.01	46.92	14
STANet-Base	12.76	70.31	11
STANet-Bam	12.85	391.17	24
STANet-Pam	13.356	512	50
TINY-CD	0.29	5.79	10
MAFGNet	6.58	62.18	125
ELGC-Net	10.57	120.69	55
LightCDNet-S	0.34	1.68	17
LightCDNet-B	1.31	3.16	22
LightCDNet-L	2.82	5.69	27
STNet	14.6	9.61	23
DDLNet	12.67	7.35	14
BAN	68.88	29.21	23
DDPM-CD	39.08	742.48	407
MambaCD-Base	85.53	89.66	65
DSFI-CD	10.135	22.28	46

the image pairs are less due to their number. Therefore, each SOTA method cannot be trained well on this dataset, so it is not as effective in this dataset as other datasets.

However, when compared with other models, DSFI-CD performs remarkably well in these settings, and we think this is because of its superior model architecture. Fig. 6(a)–(d) displays the four large-scale CD image pairs that we have chosen from the test set of CLCD. It is evident that DSFI-CD performs remarkably well in terms of overall detection accuracy as well as preventing these phenomena from happening in scenarios where other methods are prone to misdetections and omissions.

2) Visualization on CDD Dataset (Fig. 7): In contrast to DSIFN-CD, the CDD dataset yields superior detection results for all models because of its improved lighting, significant differences before and after modifications, and other factors. However, in comparison with other models, DSFI-CD continues to produce superior detection results.

As shown in Fig. 7(a)–(d), we therefore chose four sample image pairs from the CDD test set. The houses and some of the roads leading to the houses make up the majority of the changing regions in Fig. 7(a). The roads leading to the houses are obscured by trees, and the houses themselves cast shadows in the presence of light, making it more challenging to identify the boundaries of the changing regions. A building is the change zone in Fig. 7(b), but because trees partially cover the building's boundaries, it is more challenging to identify its edges. When facing Fig. 7(a) and (b), the boundary information can be well processed thanks to the special EE module in DSFI-CD, resulting in enhanced detection accuracy. Although the significant difference in light levels before and after the change in Fig. 7(c) and (d) provides a favorable environment for CD, other models are more likely to misidentify or overlook the change because of the more intricate boundaries of the change area. To some extent, DSFI-CD addresses this issue.

3) Visualization on WHU-CD Dataset (Fig. 8): WHU-CD is a building CD dataset with optimal lighting circumstances, and we chose four pre- and post-change image pairings from

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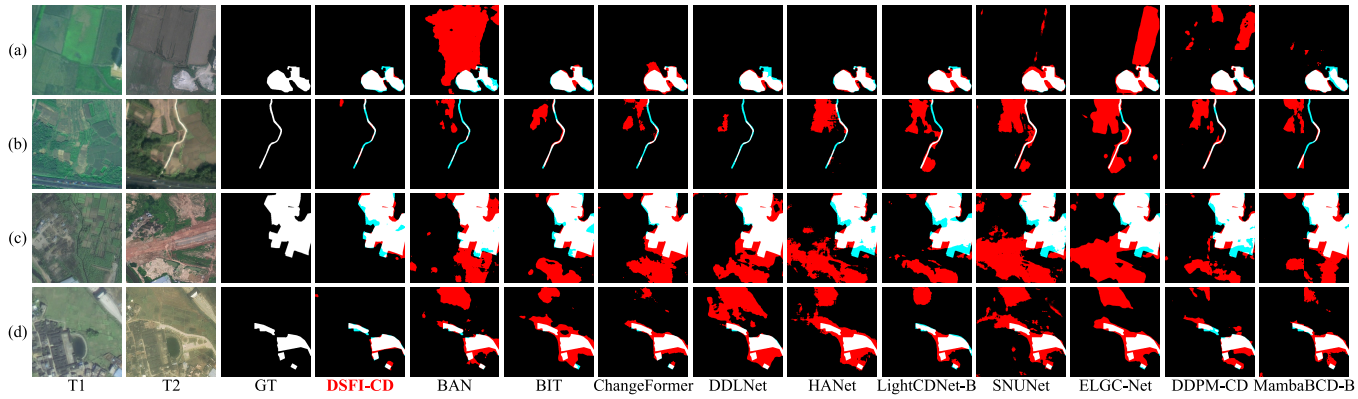


Fig. 6. Qualitative visualization of the evaluation results of DSFI-CD versus other methods on the CLCD dataset. The white and black parts of each image indicate the correctly detected change and nonchange regions, respectively, and the red and cyan indicate FP and FN, respectively. (a)–(d) Four pairs of images randomly chosen from the dataset.

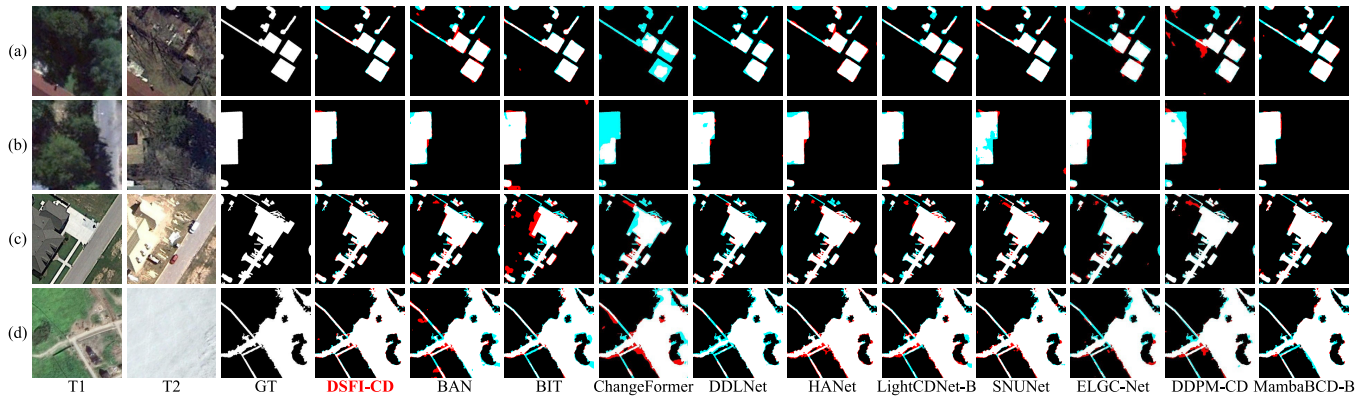


Fig. 7. Qualitative visualization of the evaluation results of DSFI-CD versus other methods on the CDD dataset. The white and black parts of each image indicate the correctly detected change and nonchange regions, respectively, and the red and cyan indicate FP and FN, respectively. (a)–(d) Four pairs of images randomly chosen from the dataset.

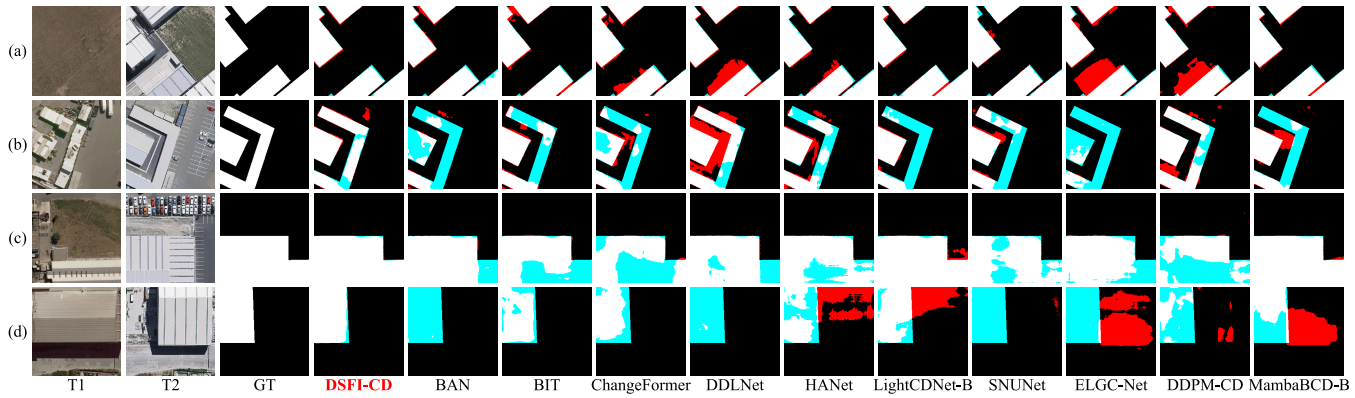


Fig. 8. Qualitative visualization of the evaluation results of DSFI-CD versus other methods on the WHU-CD dataset. The white and black parts of each image indicate the correctly detected change and nonchange regions, respectively, and the red and cyan indicate FP and FN, respectively. (a)–(d) Four pairs of images randomly chosen from the dataset.

the test set to demonstrate DSFI-CD's superiority over other models, as shown in Fig. 8.

In Fig. 8(a), the prechange image is an open field, and the postchange image is a group of houses, which is easy to identify. However, in the postchange, the houses are close to each other because some models are prone to sticking between nearby buildings. For example, in the lower right corner of the image, sticking happens in the BIT, DDLNet, LightCDNet-B, and DDPM-CD; in the upper left corner of the image, BAN, and BIT appear to be affected by this phenomenon. Pseudoconomies, or nonbuilding alterations, also happened in the images in addition to the aforementioned abnormalities. DSFI-CD successfully prevented above issue.

The buildings in Fig. 8(b) and (c) are the postchange image, while the storage warehouses are in the prechange image. Although some models interpret these sections as unmodified,

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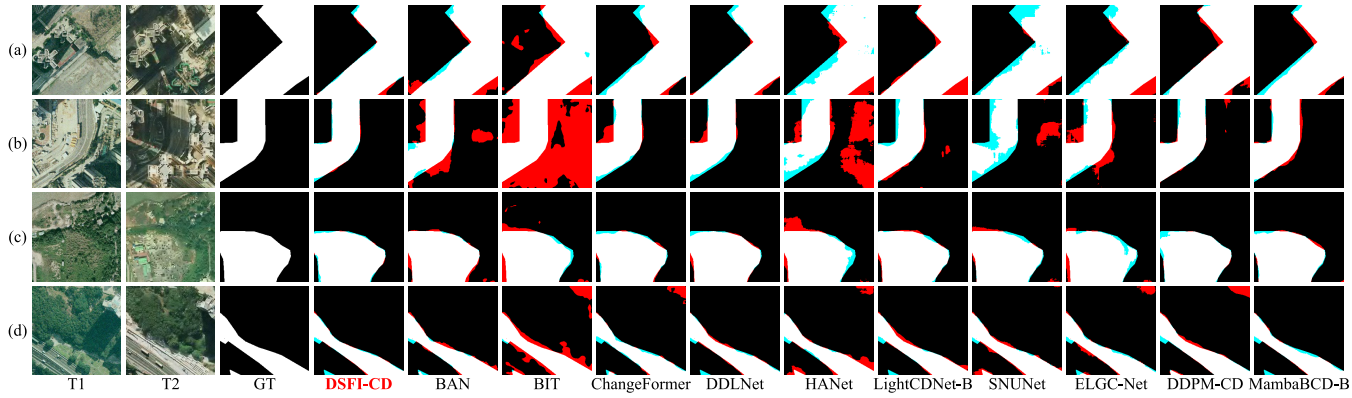


Fig. 9. Qualitative visualization of the evaluation results of DSFI-CD versus other methods on the SYSU-CD dataset. The white and black parts of each image indicate the correctly detected change and nonchange regions, respectively, and the red and cyan indicate FP and FN, respectively. (a)–(d) Four pairs of images randomly chosen from the dataset.

the region in question should be identified as altered. The buildings in Fig. 8(d) are all on the upper side of the image before the change, but after the change, they are only on the right side of the higher side. The left side of the upper side should be the change region, but some models treat all regions above as change regions.

Since DSFI-CD acts interactively on both the spatial and frequency domains, we believe that it can efficiently recognize frequency-domain elements in an image and fuse them into the spatial domain, resulting in more effective semantic information and more accurate outcomes.

4) Visualization on SYSU-CD Dataset (Fig. 9): In order to confirm the model's generalizability, we tested it on SYSU-CD with multicategory variants in this study. As with other datasets, DSFI-CD performs exceptionally well, which is illustrated in Fig. 9(a)–(d).

DSFI-CD outperforms other SOTA models on SYSU-CD primarily in terms of change area boundary detection accuracy. Unlike WHU-CD, change types in SYSU-CD include not only buildings but also highways and vegetation, making the change area boundaries difficult to see despite the good lighting conditions. The change in Fig. 9(a) from open space to a construction site is evident; yet, in certain places, the tilt photography-caused shadows of the structures made it difficult for other SOTA models to detect the change in position. Better illumination can be seen from Fig. 9(c), where the site has been altered, albeit it is harder to determine exactly, where the boundaries have been drawn. In the aforementioned scenarios, DSFI-CD produces the greatest outcomes. Most models can get good detection results for the road changes in Fig. 9(b) and (d), but DSFI-CD can still produce better results.

F. Ablation Studies

To fully validate the impact of each module in the model on model performance, we carried out ablation studies, as shown in Tables IV–VI.

1) Effects of Different Modules in DSFI-CD: To verify the impact of different modules in DSFI-CD on the overall performance of the model, we added the JD module, SFI module, and EE module in different combinations to the baseline and conducted experiments on the WHU-CD dataset, with the results shown in Table IV. It is evident that on each dataset,

TABLE IV
ABLATION STUDY OF DIFFERENT MODULES ON THE WHU-CD DATASET

Module			WHU-CD					
JD	SFI	EE	F1	Pre	Rec	IoU	$\mathcal{K}_c$	OA
Baseline			92.87	95.65	90.24	86.68	92.51	99.32
✓	✗	✗	94.07	95.74	92.45	88.8	93.77	99.43
✗	✓	✗	93.22	93.97	92.47	87.29	92.87	99.34
✗	✗	✓	93.15	92.52	**93.77**	87.17	92.79	99.32
✓	✓	✗	**94.27**	95.67	**92.91**	**89.16**	**93.98**	**99.44**
✓	✗	✓	**94.19**	**95.98**	92.47	**89.02**	**93.89**	**99.44**
✗	✓	✓	93.42	**96.65**	90.4	87.65	93.09	99.37
DSFI-CD			**94.5**	**95.82**	**93.22**	**89.58**	**94.22**	**99.47**

the model's detection performance improves when additional modules are added in comparison to the baseline. Similarly, when additional modules are added in comparison to the combination of only one module, the detection performance of the combination of modules improves. It is worth noticing that the detection accuracy of adding a JD module is highest when only one module is introduced, which we believe is related to the fact that the JD module is added at each stage of the backbone, as we will verify in Section IV-F2.

2) Effects of Different Stages of JD Module: Given that JD module is added to each stage of backbone in DSFI-CD, we evaluated the effect of adding different layers of JD module on model identification performance, and the findings are displayed in Table V. The row DSFI-CD w/o JD module in Table V indicates that the JD module is not added in each stage, which corresponds to the addition of the SFI module and EE module on the top of baseline in Table IV. It is obvious that the detection accuracy of DSFI-CD grows sequentially with the number of layers added to the JD module, demonstrating that the JD module can handle semantic information from multiscale characteristics in backbone well. Furthermore, we find that the improvement when the JD module is added only in the first stage compared with no JD module is greater than when the JD module is added in subsequent stages, which we believe is due to the fact that the spatial resolution of the feature maps is much higher in the initial stage, allowing JD module to capture a large amount of spatial texture detail information.

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TABLE V
ABLATION STUDY OF DIFFERENT STAGES ON JD MODULE IN THE WHU-CD DATASET

Model	S1	S2	S3	S4	WHU-CD F1	Pre	Rec	IoU	$\mathcal{K}_c$	OA
DSFI-CD w/o JD Module	✗	✗	✗	✗	93.42	**96.65**	90.4	87.65	93.09	99.37
DSFI-CD	✓	✗	✗	✗	94.15	95.74	92.6	88.94	93.85	**99.43**
DSFI-CD	✓	✓	✗	✗	**94.17**	95.26	**93.09**	**88.98**	**93.87**	**99.43**
DSFI-CD	✓	✓	✓	✗	**94.4**	**95.97**	**92.88**	**89.39**	**94.11**	**99.46**
DSFI-CD	✓	✓	✓	✓	**94.5**	**95.82**	**93.22**	**89.58**	**94.22**	**99.47**

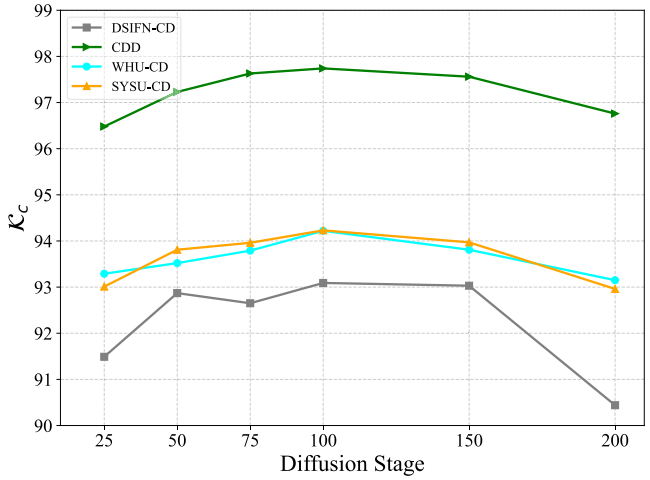


Fig. 10. Impact on $\mathcal{K}_c$ of various diffusion stages for each datasets.

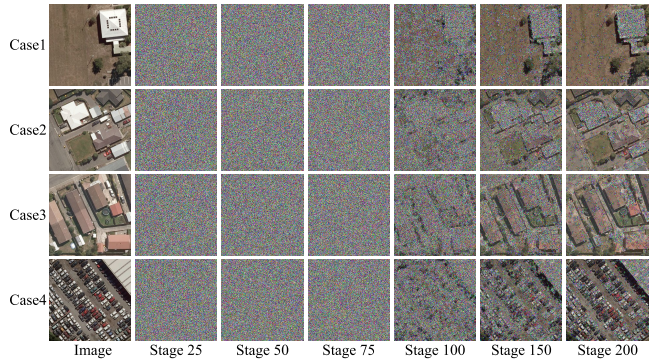


Fig. 11. Diffusion-generated samples from selected images in the WHU-CD dataset.

TABLE VI
MEMORY CONSUMPTION AND FPS FOR DIFFERENT DIFFUSION STAGES

Diffusion Stage	Memory Consumption (GB)	FPS
25	3.24	83.56
50	6.47	60.63
75	9.69	46.95
100	12.91	38.80
150	15.49	33.86
200	19.36	28.43

3) Effects of Different Diffusion Stages in JD Module: In a vast number of DDPMs research, diffusion stages are frequently a critical statistic that reflects the extent to which noise is decreased during model diffusion. In this article, these values of [25, 50, 75, 100, 150, 200] are used as the diffusion stages in the experiments, and the experimental results on the four datasets are shown in Fig. 10. In addition, we visualized some of the pseudoimages generated by different diffusion steps and the memory consumption required for sampling different diffusion steps and calculated their FPS in Fig. 11 and Table VI, respectively. It can be seen from the above results that with the increase of the diffusion step, the sampling process demands more and more memory, and at the same time the generated pseudoimages contain more semantic information, but when the diffusion step is too large then more information in the original image will be lost, and the final results show that when the diffusion step is chosen to be 100, it can achieve a better balance, and it is optimal in terms of the evaluation index $\mathcal{K}_c$.

4) Effects of Different Backbones in Encoder: We also simply compare the MiT-B0 with some representative backbone networks in CD tasks on the CLCD dataset in Table VII. It can be seen that MiT-B0 outperforms the other MiT backbone networks and ResNet-18.

We believe that is the reason for this phenomenon. To begin, MiT-B0 outperforms ResNet-18 in feature extraction, due to primary Transformer architecture and self-attention mechanism, which can effectively capture global contextual information and long-distance dependencies, whereas ResNet-18 relies on convolutional operations with a limited perception field, making it difficult to deal with complex global relationships. Furthermore, MiT-B0 achieves multiscale feature extraction through the layered design, merging local and global information, while its dynamic weight allocation mechanism can be flexibly altered according to the inputs to further boost the feature expression potential. Second, MiT-B0 outperforms MiT-B1 and MiT-B2 in certain tasks, due to its lightweight design, which is better suited for basic tasks, limited datasets, or resource-constrained scenarios with low optimization complexity and danger of overfitting. MiT-B1/B2, while more powerful, may underperform in some cases due to model complexity, inadequate data, or resource constraints.

G. Network Visualization

To present the feature maps at different phases of the inference process more intuitively, we employ Grad-CAM [96] to visualize DSFI-CD, and the results are given in Fig. 12.

We selected three sets of images from the WHU-CD dataset that show the before and after change; one can observe that

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TABLE VII
ABLATION STUDY OF DIFFERENT BACKBONE TYPES ON THE CLCD DATASET

Encoder	F1	Pre	Rec	IoU	$\mathcal{K}_c$	OA	Param(M)	GFLOPs
MiT-B0 [79]	**78.94**	**84.74**	**73.88**	**65.2**	**77.37**	**97.07**	10.14	22.28
MiT-B1 [79]	**75.88**	**79.93**	**72.21**	**61.13**	**74.04**	**96.58**	37.1	38.36
MiT-B2 [79]	**70.58**	71.54	**69.64**	**54.53**	**68.25**	**95.68**	48.15	38.48
ResNet-18 [30]	59.8	**76.04**	49.28	42.66	57.3	95.07	32.96	70.46

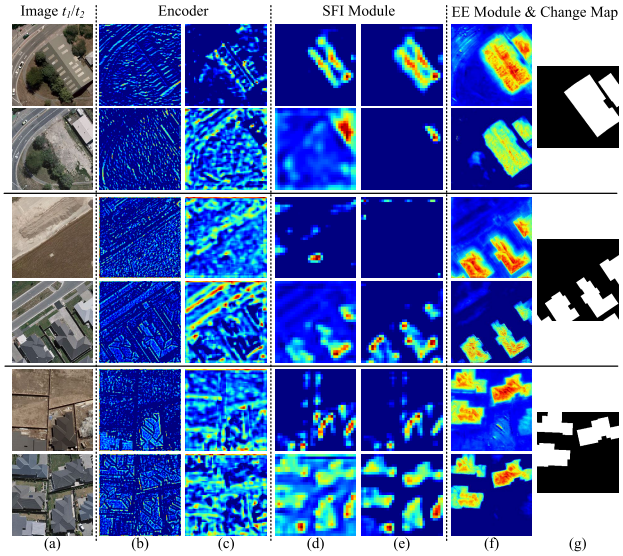


Fig. 12. Visualization of different layer features in DSFI-CD. (a) Input images. (b) and (c) Stages 2 and 3 features in encoder. (d) and (e) Features before and after inputting the SFI module. (f) Features from EE module. (g) Change map.

the model moves from extracting detailed texture features in the original image to extracting deeper semantic features with an increase in the number of layers. Column (a) lists the before and after change images, and Columns (b) and (c) list the encoder intermediate Stage 2 and Stage 3 features; it is evident from Columns (d) and (e) that the model suppresses the effects of irrelevant regions while capturing more effective characteristics through the use of spatial–frequency domain features in the SFI module; Column (f) presents feature maps with an up-and-down arrangement of the boundary information extracted by the EE module before and after. It is evident that employing EE module It is evident that while the model was able to extract the changing areas in crude form prior to the use of EE module, the latter allowed for even more boundary refinement while suppressing features in nonchanging regions.

V. CONCLUSION

In this article, we propose a new CD method DSFI-CD, in which we first use the excellent image generation capability of conditional DDPM to indirectly generate pseudovariable image pairs, and then use the JD module to fuse these pseudovariable image pairs with the original images, and then enhance the model's feature extraction capability in encoder. In the decoder section, first, we create the SFI module for the characteristics from the encoder in the first place. Through these two branches, we extract the features' frequency-domain information and their spatial domain information, respectively, and we use them to extract the features' information comprehensively. Second, we design the EE module to further refine the features' information about the boundary of the changing region. We use operations like multilevel feature fusion and the Laplace pyramid to improve the sensitivity of the model to the changing region's boundary while reducing the interference of the nonchanging region.

In Section IV, we conduct a series of comparative experiments on four publicly available CD datasets to validate the advantages of DSFI-CD over other SOTA models, followed by ablation experiments on these datasets to validate the impact of the modules in DSFI-CD on the model's performance and visualize the model's feature at each stage of the inference process.

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